CSE 216 Formal Methods
Fall 2026 Daniel Fremont

HW 3

Due October 20 at 3:00 pm

(4 questions, 185 points total)

1. (20 pts.) Resolution

(a) Which of the following clauses can be resolved with each other, and what are the resulting clauses?

(A) $x \vee y \vee z$
(B) y
(C) $\neg y$
(D) $\neg y \vee z$

Answer:

(a) clauses A,C, results in resolution of $x \vee z$
(b) clauses A,D results in $x \vee z$
(c) clauses B,C results in $\bot$
(d) clauses B,D results in z

(b) Give a resolution refutation for the UNSAT formula with the following clauses:

$$c_1 = \neg x \vee \neg y \vee z$$
$$c_2 = \neg x \vee w$$
$$c_3 = x \vee \neg y$$
$$c_4 = \neg y \vee \neg z \vee \neg w$$
$$c_5 = x \vee y$$
$$c_6 = y \vee \neg w$$

You may write $c_i \odot c_j$ to denote the resolution of two clauses.

Answer:

i. $c_7 = c_3 \odot c_5 = (x \vee \neg y) \odot (x \vee y) = x$
ii. $c_8 = c_2 \odot c_7 = (\neg x \vee w) \odot (x) = w$
iii. $c_9 = c_6 \odot c_8 = (y \vee \neg w) \odot (w) = y$
iv. $c_{10} = c_1 \odot c_7 = (\neg x \vee \neg y \vee z) \odot (x) = \neg y \vee z$
v. $c_{11} = c_4 \odot c_9 = (\neg y \vee \neg z \vee \neg w) \odot (y) = \neg z \vee \neg w$
vi. $c_{12} = c_{11} \odot c_8 = (\neg z \vee \neg w) \odot (w) = \neg z$
vii. $c_{13} = c_1 \odot c_{12} = (\neg x \vee \neg y \vee z) \odot (\neg z) = \neg x \vee \neg y$
viii. $c_{14} = c_{13} \odot c_7 = (\neg x \vee \neg y) \odot (x) = \neg y$
ix. $c_{15} = c_{14} \odot c_9 = (\neg y) \odot (y) = \bot$

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2. (60 pts.) CDCL

Consider running CDCL on the formula in (1b) above, with a variable selection heuristic that prefers $z > w > x > y$ and a phase selection heuristic that always picks the negative polarity.

(a) Up to the first conflict, what decisions are made?

Answer: z is false at DL1

w=false at DL2

x is false from c_2 and then y is false from c_3. this leads to c_5 being unsat.

(b) Construct the implication graph for the first conflict.

Answer: see figure 1

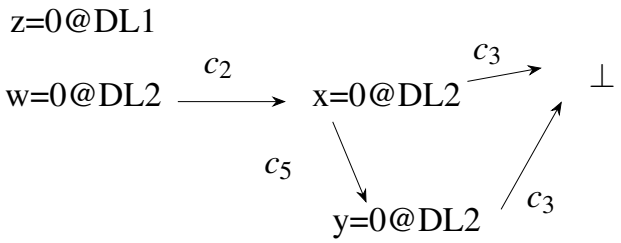


Figure 1: 2b

(c) What are the possible conflict clauses that could be learned from this conflict? Which of these is asserting if we backtrack one level?

Answer: we could learn

(a) $x \vee y$

(b) x

(c) w

we can say that w must be true if we go back to DL1.

(d) If we use the first UIP heuristic, which conflict clause will CDCL learn, and what decision level will it backtrack to?

Answer: the conflict clause learned is x. it will backtrack to DL0 since the "second highest" is the empty clause which is at DL0.

(e) Construct the implication graph for the next conflict.

Answer: see fig 2

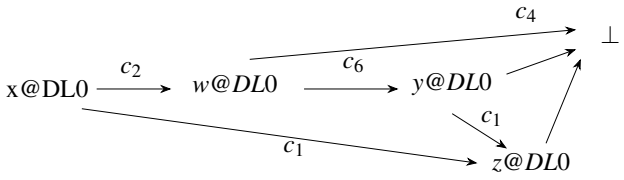


Figure 2: 2e

(f) CDCL terminates after these two conflicts. Why? How can you use the implication graphs from each conflict to build a resolution refutation of the original formula?

Answer: the second conflict happens at DL0 which means the formula is unsat.

$c_3(x \vee y)$ resolved with $c_5(x \vee y)$ on y gives x.

c_2, c_6, c_1, c_4 with x resolve to $\perp$

3. (70 pts.) First-order logic

For each of the following formulas, state whether they are satisfiable or valid (or neither). If a formula ϕ is satisfiable, give a model, i.e. a structure M such that $M \models \phi$. If ϕ is not valid, give a structure M such that $M \not\models \phi$. For each formula, assume the signature of the logic — the constant, function, and relation symbols — consists just of those symbols used in the formula. We use c, f, and R as constant, function, and relation symbols respectively.

(a) $\forall x.\forall y.(f(x,y) = f(y,x))$

Answer:

(b) $\exists x.(x = x)$ *(hint: can you pick the universe of the structure so that this is false?)*

Answer:

(c) $\forall x.\forall y.(x = y)$

Answer:

(d) $\exists x.(x = x) \wedge \forall x.\forall y.\neg(x = y)$

Answer:

(e) $\forall x.\exists y.(f(x) = y)$

Answer:

(f) $\exists x.(R(x) \wedge \neg R(f(x)))$

Answer:

(g) $[R(c) \wedge \forall x.(R(x) \rightarrow R(f(x)))] \rightarrow R(f(f(f(c))))$

Answer:

4. (35 pts.) Arrays

Consider the theory of bitvectors (say, 32-bit) and arrays whose indices and values are bitvectors (sometimes called ABV). This theory has two kinds of objects, or *sorts*: bitvectors and arrays. Let's use the notation $\forall x : Array.$ and $\forall x : BV.$ to quantify over arrays and bitvectors respectively. So for example if we use the C notation $>>$ for the right shift operator, then $\forall x : BV.(x >> 32 = 0)$ would be a valid formula asserting that for all 32-bit bitvectors, right shifting by 32 bits yields the zero bitvector.

(a) Write an SMT formula $\phi(x)$ which is true if and only if the array x is sorted in increasing order.

Answer:

(b) Write a formula $\phi(x)$ which asserts that every element of the array x has the same value.

Answer:

(c) Write a formula with no free variables (a *sentence*) asserting that if element 0 of an array is even, and for every i if element i is even then element $i+1$ is even, then all elements of the array are even. Is this formula valid?

Answer:

5. (optional, bonus problem; 0 pts.) GCHQ puzzle

The British signals intelligence organization GCHQ released a puzzle in December 2015, the correct solution of which led to a sequence of increasingly difficult puzzles (presumably with your employment as a codebreaker at the end — alas, the puzzles have been taken down). They described it as "a brain-teasing puzzle that seems certain to exercise the grey matter of participants over the holiday season". Don't exert yourself: solve it with SAT.